

1. This musical element is influenced by how many instruments are playing and how many different vocal or instrumental parts are in a piece. It might be described as thick or thin and shares its name with an element of art. Name this element concerned with the relationships between a musical piece's layers.

TEXTURE

2. Identify this number by its definition. This integer is a real number and a whole number larger than negative one. What is this integer that is not a natural or counting number?

0 (ZERO)

3. This fantasy novel is set in the Kingdom of Dor, which is ruled by Princess Pea's father. At its conclusion, the princess, Miggery Sow, Roscuro [ro-SKYOOR-oh] the rat, and the title mouse all reach their own happy endings. Give the title of this Newbery-winning novel by Kate DiCamillo [dee-kuh-MILL-oh].

(THE) TALE OF DESPEREAUX [DES-puh-row]

4. The universities of Brescia and Kentucky Wesleyan are located in this city, Kentucky's fourth-largest by population. Identify this Kentucky city bordering Indiana which was the birthplace and childhood home of actor Johnny Depp.

OWENSBORO

5. When this chemical reaction happens very quickly, an explosion can result. It begins when a substance reaches its ignition point, usually by being struck by a spark, exposed to strong light, or heated. Fire is the most common type of what chemical reaction that produces heat and light?

COMBUSTION

6. The vendor had 72 keychains for sale. He sold all of the keychains. If each customer bought 3 keychains, how many customers bought the keychains?

24 (TWENTY-FOUR CUSTOMERS)

7. This Caldecott-winning book's last update included George W. Bush. Its subjects are the first 42 leaders of the United States. Give the title of this book written by Judith St. George and illustrated by David Small.

SO YOU WANT TO BE PRESIDENT?

8. One of the most important political figures in Kentucky history, he served as both Speaker of the House and Secretary of State. He was also an influential U.S. senator, multi-time candidate for president, and a leading member of the Whig party. Who was this political figure often referred to as the "Great Compromiser"?

(HENRY) CLAY

9. Originally, this term referred to hymns that celebrated the birth of Jesus. Later, it spread to songs about snow and Santa Claus. What do we call these seasonal Christmas songs?

(CHRISTMAS) CAROL(S)

10. Confusingly, this force isn't really a force, but instead is the result of inertia [ih-NUR-shuh]. A tetherball exhibits it, as does a person riding a merry-go-round. Name this force, which is the energy of an object trying to move in a straight line when it's actually moving in a circle.

CENTRIFUGAL (FORCE)

11. This phenomenon occurs when a light, sound, or other wave encounters a small object in its path. It also happens when the wave is forced through a small opening. Which physics process happens when a wave remains in the same medium but bends around an object?

DIFFRACTION

12. In both World Wars, this instrument accompanied the Highland Regiment into battle. It has a drone and a chanter attached to a sack that, in earlier centuries, was made of animal skin. Name this traditional Scottish wind instrument.

BAGPIPE(S)

13. This type of water body is generally larger than a bay, but typically features a narrower opening. Give the name for these large bodies of water, an example of which borders Florida to the west and the states of Mississippi, Louisiana, and Alabama to the south.

GULF

14. Besides rhyme, this sound device is also part of the sentence, "It's true--I do like you!" What is the name for this repetition of vowel sounds?

ASSONANCE

15. The garden path is 15 feet long. The path is covered with 20 equal-sized stones. As a fraction in lowest terms and in feet, how big is each stone?

3/4 (THREE-FOURTHS FOOT)

16. In 2012, Max Richter released a so-called "recomposed" version of this Baroque violin suite. The original works were published with accompanying poems that the composer may or may not have written. What is the collective title for this Antonio Vivaldi work consisting of "Spring," "Summer," "Autumn," and "Winter"?

(THE) FOUR SEASONS

17. This author's first novel, *A Summer to Die*, was based on the loss of her sister. She also created the fictional siblings Anastasia and Sam Krupnik. Give the name of this author of the *Gooney Bird Greene* series.

(LOIS) LOWRY

18. This part of soil helps form pores and keeps the soil loose and soft. It also provides the nutrients that plants need. Identify this organic matter made from decaying plants and animals.

HUMUS [HYOO-mus]

19. Three-word answer required. An American family's communication by way of this language formed the plot of the 2022 Oscar winner for Best Picture, *CODA*. This visual-based, natural language is the primary form of language for hundreds of thousands of Americans within the Deaf communities. Identify this form of language often abbreviated ASL.

AMERICAN SIGN LANGUAGE

20. Seventy-five percent of the theater seats were filled. There are 200 seats in the theater. How many seats were filled?

150 (ONE HUNDRED FIFTY SEATS)

21. Essentially, this literary technique is the opposite of a flashback. In it, an author may point out objects that will prove important or describe an event similar to what will happen. Identify this type of spoiler that hints at what will occur later in a work.

FORESHADOW(ING)

22. Some researchers have suggested the single letter within the name of this generational group could stand for "zoomers". Give the name for the generation of individuals, now roughly between the ages of 10 and 25, which directly followed a generation known as the Millennials.

(GEN(ERATION)) Z

23. This composer designed the Bayreuth [by-ROIT] Festival Theater in Bavaria to house his operas. Those works include *The Flying Dutchman*, *Parsifal* [PAR-sif-uhl], and *Lohengrin* [LO-hen-grin]. Whose masterpiece was the four-opera *Ring Cycle*?

(RICHARD) WAGNER [RIKH-ahrt VAHG-nuhr]

24. This atmospheric layer extends to about 30 miles above Earth's surface. It lies above the troposphere [TRO-pus-feer] and below the mesosphere [MEZ-uh-sfeer]. What is this home of the ozone layer?

STRATOSPHERE

25. The gym teacher set up an obstacle course at school. Students have to bounce a ball for the first 6 feet, then run 12 feet to the end of the field, and finally hop back to the beginning of the course. How many yards must the students hop?

6 (SIX YARDS)

26. A regular pentagon has five equal interior angles. What is the measure of an interior angle of a regular pentagon in degrees?

108 (DEGREES)

27. This fable ends with a group of insects dismissing another insect's request for food. The starving bug spent the summer playing music while his neighbors stored grain for winter. Identify this Aesop fable teaching that, quote, "There's a time for work and a time for play."

(THE) ANT(S) AND THE GRASSHOPPER

28. Residents of Ohio are often referred to by this term which is also the mascot for The Ohio State University. Give this term, which can describe a type of peanut butter candy dipped in chocolate, found in Ohio's state nickname.

BUCKEYE(S)

29. This 19th-century social dance possibly originated in Germany before spreading to France and England. It required a couple to perform a slide, close, slide step in 2/4 time. What was this dance, whose name sounds like a horse's run?

GALOP [GAL-up]

30. Scientists first discovered these so-called "light clocks" in 1967 and, to date, have found about 2,000. These rapidly spinning neutron stars can emit radio waves, visible light, x-rays, or gamma rays. Identify these stars named for the regular bursts of energy they emit.

PULSAR(S)

End of First Half – Begin Three-Minute Inquiry Period

Round 4 - Second Half

Elementary – 2023 Region



31. Like the endoplasmic reticulum [in-doh-PLAZ-mik ruh-TIK-yuh-lum], this organelle resembles a stack of pancakes. It absorbs transport vesicles from the E.R. and forms lysosomes [LIE-suh-zomes]. Identify this organelle that packages proteins for transport in vesicles.

GOLGI [GOAL-jee] APPARATUS (Accept GOLGI BODY, GOLGI COMPLEX)

32. This man won Emmy Awards for directing and choreographing the TV special *Liza with a Z*. His record eight Best Choreography Tonys came for musicals like *Pippin*, *Sweet Charity*, and *The Pajama Game*. Who directed and co-wrote the semi-autobiographical movie *All That Jazz*?

(BOB) FOSSE [FOSS-ee]

33. This fairy tale inspired the title of a 1941 A. A. Milne play. Since its publication, that title has become a saying for, quote, "an unattractive or unpromising child who becomes a beautiful or much-admired adult." Identify this Hans Christian Andersen fairy tale about an unloved young bird who becomes a beautiful swan.

(THE) UGLY DUCKLING

34. Three-word answer required. Playwright Arthur Miller famously depicted these events in his Tony Award-winning play, *The Crucible*. Some 200 residents of colonial Massachusetts were accused of witchcraft in what controversial series of prosecutions during the late 1600's?

SALEM WITCH TRIALS

35. Give your answer in terms of pi. The diameter of a circle is 6 units. What is the circumference of the circle in units?

6 PI (UNITS)

36. This theatrical form originated in 17th-century Kyoto. Like noh, it requires performers to act, sing, and dance. Heavy make-up and all-male casts are features of which type of Japanese drama?

KABUKI [kuh-BOO-kee]

37. This suffix meaning "condition of" or "skill" often means a word is a noun. It may follow "relation," "leader," "friend," or "champion." Identify this suffix that, as a standalone word, names a large boat.

SHIP (Accept stated or spelled)

38. A spinner is divided such that one-sixth of the sectors are red, one-half of the sectors are green, and one-third of the sectors are white. On which color is the spinner least likely to land?

RED

39. These plant parts protect and enclose a bud as it develops into a flower. What are these outermost flower organs that resemble small leaves?

SEPAL(S)

40. Delivered at the dedication of the Soldiers' National Cemetery on November 19, 1863, this brief speech began with the words "Four score and seven years ago". Give the name of this iconic speech delivered during the American Civil War by President Abraham Lincoln.

GETTYSBURG ADDRESS

41. This word can be a synonym of "student." Identify this word that also names the black, central portion of the eye.

PUPIL

42. From 1817 to 1825, he served as the United States' fifth president. Which early American president is best remembered today for his "doctrine" warning European powers against taking any further action in the Western Hemisphere?

(JAMES) MONROE

43. You want to make as many different outfits as possible from the clothes in your closet. You have 6 shirts, 5 bow ties, and 2 pairs of pants in the closet. How many outfits can you make from these different clothes?

60 (SIXTY OUTFITS)

44. Organisms with this type of symmetry don't have left or right sides but do have fronts and backs. They can repeat a body part 4, 5, 6, or 8 times around a central axis. Identify this symmetry exhibited by the sea anemone [uh-NEM-uh-nee] and the starfish.

RADIAL (SYMMETRY)

45. Ancient artists created this type of art with colored pebbles. Its pieces are called tesserae [TESS-uh-ree], which means "dice" or "cubes" in Latin. Identify this long-lasting art form that consists of small glass, ceramic, or stone pieces embedded in plaster.

MOSAIC

46. A geometric sequence begins with 14, 28, and 56. What is the fifth term in the sequence?

224 (TWO HUNDRED TWENTY-FOUR)

47. This art style believed in letting the viewer's eye mix colors instead of having an artist mix them on a palette or canvas. Its pioneers were the French painters Paul Signac [seen-YAK] and Georges Seurat [ZHWARZH suh-RAH]. Identify this movement named for the small dots of color it favored over brushstrokes.

POINTILLISM (Accept word forms)

48. In high temperatures, this animal's short-beaked variety cools itself by blowing mucus bubbles that burst and evaporate. Also called the spiny anteater, it resembles--but is unrelated to--the hedgehog. Give the name of this species that, with the platypus, is Australia's only monotreme.

ECHIDNA(S) [uh-KID-nuh]

49. A major road within Lexington, Kentucky is named for this legendary racehorse who won 20 of the 21 races run during his career. Fittingly, this thoroughbred's only loss came to a horse named Upset. Give the name of this animal widely often considered the horse racing's all-time greatest champion.

MAN O' WAR

50. These paired punctuation marks may enclose a side comment or side sentence. They show that the information between them is a non-essential clause. Name these punctuation marks that look like curved brackets.

PARENTHESES

51. Let D represent the number of dogs in the pet store. Let C represent the number of cats in the store. What equation shows that there are a total of 16 dogs and cats in the store?

$D + C = 16$ (D PLUS C EQUALS 16) (ACCEPT: $C + D = 16$)

52. This group of about 30 paintings was the artist's first series dedicated to a single subject. Completed in 1890 and '91, it examined seasonal light changes on piles of a feed crop for horses and cattle. Identify this Monet series.

HAYSTACKS

53. This three-word title is among those held by the president of the United States and refers to the president's power within the U.S. Armed Forces. What title gives the president absolute control and authority of the military?

COMMANDER IN CHIEF

54. This body system's major organs include the thymus, spleen, appendix, and tonsils. A relative of the circulatory and immune systems, it is the reason bone marrow produces white blood cells. What is this system named for a yellowish fluid that small vessels transfer from the body's tissues to the bloodstream?

LYMPHATIC (SYSTEM) (Accept LYMPHOID (SYSTEM))

55. What is the following sentence's direct object? "For my 11th birthday, my dad built me a clubhouse."

(A) CLUBHOUSE

56. Identify the following sentence's appositive phrase: "A record-setting runner, Shakira practically lived at the track."

A RECORD-SETTING RUNNER

57. This type of data is either discrete or continuous. Those words mean the data is either countable or measurable. State this opposite of qualitative data.

QUANTITATIVE (DATA)

58. Sharon painted 22 roses and 6 daisies on the mural. What is the ratio of daisies to the total number of flowers that Sharon painted?

3:14 (THREE TO FOURTEEN) (ACCEPT: 3/14 OR THREE OVER FOURTEEN)

59. Economists often use a graph to determine the existence of this economic situation by pinpointing the moment that an item's supply and demand meet. Give the name for this moment in a market in which the available supply of a good or service is perfectly balanced with the current demand for it.

(MARKET) EQUILIBRIUM

60. In the mid-1800's, women sometimes held so-called "bees" to complete this type of folk art. It has a layer of insulating batting tucked between two pieces of cloth. Name this kind of blanket that can be pieced or patchwork.

QUILT(S)

End of Second Half – Begin Three-Minute Inquiry Period

Extra Questions

61-AH. This United Nations Messenger of Peace was arrested for refusing to fight in the Vietnam War. He earned an Olympic gold medal at age 18 and was the first boxer to win the world heavyweight championship three times. Name this Louisville native, who claimed he could, quote, "float like a butterfly, sting like a bee."

(MUHAMMAD) ALI

62-LA. Spelling required. "He hopes to duplicate his overseas success domestically." Spell the word in the preceding sentence meaning, quote, "to produce something equal to."

D-U-P-L-I-C-A-T-E

63-MA. A bag contains 3 red cubes, 4 blue cubes, and 7 orange cubes. What is the probability of choosing an orange cube on the first draw?

1/2 (ONE-HALF) (ACCEPT: 50 PERCENT)

64-SC. This physician used hypnosis to help his patients speak freely about their lives. He divided the human mind into the id, ego, and superego and wrote *The Interpretation of Dreams*. Identify this Viennese man nicknamed the "father of psychoanalysis."

(SIGMUND) FREUD [FROYD]

65-SS. In a popular musical parody concerning this type of digital asset, rapper Jack Harlow explains, "its unique, there can only be one". In December 2022, former president Donald Trump announced he was selling a line of them featuring artistic portraits of himself. Identify these blockchain-based assets, often produced in the form of photos or videos, that are often bought and sold under a three-letter abbreviation.

NFT (ACCEPT: NON-FUNGIBLE TOKEN)